

RESEARCH STATEMENT

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I work in algebraic combinatorics, using the combinatorics of reflection groups to study algebraic varieties and their coordinates. My main tool is the *affine Deodhar decomposition*: a partition into pieces of the form $(\mathbb{C}^*)^r \times \mathbb{C}^s$, products of algebraic tori and affine spaces, where $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$. I introduced periodic strand diagrams in which strands cross or turn at elbows, called *affine Deograms*, to index these pieces [11]. This decomposition connects my two research programs: bijections in rational Catalan combinatorics and cluster coordinates in affine type A .

The geometric starting point is the *Grassmannian* $\text{Gr}(k, n)$, the space of k -dimensional subspaces of $\mathbb{C}^n$, represented by rank- k $k \times n$ matrices modulo row operations. Its *Plücker coordinates* Δ_I are the maximal minors indexed by k -subsets of $[n] = \{1, \dots, n\}$. An *open positroid variety* Π_f° prescribes the ranks of cyclically consecutive collections of columns [10]. Its index is a *bounded affine permutation*, a bijection $f: \mathbb{Z} \rightarrow \mathbb{Z}$ with $f(i+n) = f(i) + n$, $i \leq f(i) \leq i+n$, and average displacement k .

1. RATIONAL COXETER–CATALAN COMBINATORICS

Galashin and Lam associate to Π_f° a *positroid Catalan number* C_f , obtained by normalizing its finite-field point-count polynomial and setting $q = 1$ [5]. It counts *maximal Deograms*, the diagrams with the fewest elbows. For the top cell $f_{k,n}: i \mapsto i+k$ with $\gcd(k, n) = 1$,

$$C_{f_{k,n}} = \frac{1}{n} \binom{n}{k}.$$

This also counts *rational Dyck paths*: north-and-east lattice paths from $(0,0)$ to $(n-k, k)$ staying above the joining diagonal. Galashin–Lam–Trinh–Williams used maximal Deograms and their parking counterparts to define rational noncrossing Coxeter–Catalan objects in every finite Weyl group [8]. In type A , Dyck paths give the classical nonnesting model. The families have the same size, but a bijection is needed to transfer statistics, labels, and group actions between them.

Theorem 1 (Martinez [12]). *For coprime integers $0 < k < n$, a recursively defined bijection sends maximal $f_{k,n}$ -Deograms to rational Dyck paths:*

$$\Phi_{k,n}: \text{Deo}_{f_{k,n}}^{\max} \longrightarrow \text{Dyck}_{k,n}.$$

The construction also treats more general bounded affine permutations by restricting the allowed path region. It matches a reversible reduction of the strand diagram with a decomposition of the path into smaller rational pieces. The reduction changes the Plücker chart using my affine-diagram moves, resolving the consecutive-chart problem of [11]. Thus affine geometry supplies the recursion for the bijection.

In the *Fuss–Catalan* case $n = dk + 1$, with d a positive integer, periodicity gives explicit, nonrecursive bijections [13]. One sends maximal Deograms to Dyck paths; the other sends twisted maximal Deograms, the parking counterparts in the noncrossing model, to *rational parking functions*: Dyck paths whose north steps carry the labels $1, \dots, k$, increasing up each column. These maps put the ordinary and parking theories in one framework and make the symmetric-group action, which permutes labels and then restores column order, accessible on Deograms.

I next plan a weighted Deogram model for rational q, t -Catalan numbers [6], transporting *area*, the count of unit squares wholly between path and diagonal, and *dinv*, the count of pairs of east and later north steps met by a common line parallel to the diagonal. Further steps are to extend the direct map beyond Fuss slope and test compatibility with cyclic actions and local flips.

2. AFFINE PLABIC GRAPHS AND CLUSTER STRUCTURES

2.1. Torus patches and links

For a k -subset $I \subset [n]$, consider the *Plücker patch*

$$U_{f,I} := \Pi_f^\circ \cap \{\Delta_I \neq 0\}.$$

Snider identified it with an *open Richardson variety* $\mathring{R}_{f,t_I}$, an intersection of Schubert cells for opposite triangular subgroups in the affine flag variety [16]. This flag variety parametrizes periodic chains of lattices, free rank- n $\mathbb{C}[[z]]$ -modules spanning $\mathbb{C}((z))^n$. The permutation t_I sends i to $i + n$ for $i \bmod n \in I$ and fixes every other integer i . My (f, P_I) -affine Deograms encode this patch: f specifies the strand permutation, while the lattice path P_I , with north steps indexed by I , cuts out the fundamental region [11].

The choice of I can change the geometry sharply. The top open positroid variety in $\mathrm{Gr}(2, 4)$ has dimension four and generic diagonal-torus orbits of dimension three, yet its Δ_{13} -patch is $(\mathbb{C}^*)^4$ [14]. This illustrates the difference between being an algebraic torus and having a dense orbit under the standard diagonal action. With Matthew J. Tyler, I proved an affine analogue of the finite-type torus criterion of Gorsky–Kim–Sherman–Bennett [9].

Theorem 2 (Martinez–Tyler [14]). *For $f \leq t_I$ and $d = \ell(t_I) - \ell(f)$, the following are equivalent: $U_{f,I} \cong (\mathbb{C}^*)^d$; the Bruhat interval $[f, t_I]$ contains no 2-crown; $R_{f,t_I}(q) = (q - 1)^d$; and there is a unique (f, P_I) -affine Deogram.*

Here *Bruhat order* compares reduced words in simple reflections by subword inclusion, ℓ is their length, a 2-crown is a subinterval isomorphic to the six-element Bruhat order on S_3 , and the R -polynomial counts finite-field points of the open Richardson variety. The theorem identifies patches with one Deodhar torus chart, a natural base case for constructing cluster coordinates.

The same project connects torus actions to link topology. Finite-type *braid Richardson varieties* parametrize configurations of flags—nested chains of subspaces—with relative positions prescribed by a positive braid word and an endpoint permutation. Closing the braid followed by the inverse positive braid lift of the endpoint gives an associated *link*, an embedded union of circles in S^3 . A link is *smoothly slice* if it bounds disjoint smooth disks in the four-ball, and an *unlink* if it bounds disjoint disks in S^3 .

Theorem 3 (Martinez–Tyler [14]). *A nonempty type A braid Richardson variety has a dense orbit under the standard diagonal torus if and only if its associated link is smoothly slice. For open positroid varieties, these conditions are also equivalent to the link being an unlink.*

2.2. The cluster program

A *cluster seed* consists of algebraically independent coordinate functions and a *quiver*, a directed graph encoding exchange relations. *Mutation* replaces one mutable function using an exchange relation and changes the quiver; frozen functions stay fixed. The *cluster algebra* $\mathcal{A}$ is generated by all mutation-produced variables, together with frozen variables and frozen inverses. Its *upper cluster algebra* $\mathcal{U}$ is the intersection of the Laurent polynomial rings of all seeds. Thus $\mathcal{A} = \mathcal{U}$ asks whether every function Laurent in every chart is generated by cluster variables.

In finite type, plabic graphs—planar graphs with black and white internal vertices—and their 3D analogues give cluster structures on positroid and braid varieties [4, 7]. Their affine counterparts are *double affine braid varieties* $\mathring{R}_{f,\beta}$, indexed by f and a positive double affine braid word β ; affine Richardson varieties are special cases. Bao and Ye identify their coordinate rings with upper cluster algebras in Kac–Moody type [1]. Equality with the cluster algebra remains open in general.

With Moriah Elkin and Tyler, I am building the missing diagrammatic model of seeds [3]. Our *affine 3D plabic graphs* are periodic spatial graphs with black and white vertices; local

moves generalize square moves and braid relations. Their reduced positive members are *affine Le-diagrams*, the minimal affine Deograms. We attach a candidate seed $\Sigma(G)$ whose variables are generalized minors, representation-theoretic extensions of matrix minors, and whose quiver records the exchanges induced by local moves.

We prove that these seeds are really full rank, meaning their extended exchange-matrix rows span the mutable integer lattice, and that their cluster algebras are *locally acyclic*: their spectra have finite covers by cluster localizations with acyclic mutable quivers. The moves act by mutation and by compatible rescaling in frozen variables. Muller’s theory then gives $\mathcal{A}(\Sigma(G)) = \mathcal{U}(\Sigma(G))$ [15]. What remains is to identify our upper cluster algebra with the coordinate ring, the ring of regular functions on the variety, by matching our seeds with those of Bao and Ye.

Conjecture 4 (Elkin–Martinez–Tyler [3]). *For every double affine braid variety and every affine 3D plabic graph G for it,*

$$\mathbb{C}[\mathring{R}_{f,\beta}] = \mathcal{A}(\Sigma(G)) = \mathcal{U}(\Sigma(G)).$$

Graphs representing the same variety give mutation-equivalent seeds, with mutations realized by local moves.

For β a word for t_I , the graphs flatten to *punctured plabic graphs*, bicolored graphs on a disk with one interior point removed. We prove that affine Le-diagrams are in bijection with move-equivalence classes of reduced punctured plabic graphs. Their face labels give candidate seed functions, with the puncture recording the chosen Plücker chart.

3. FUTURE DIRECTIONS AND UNDERGRADUATE RESEARCH

My longer-term goal is to transfer structures on positroid varieties to their patches, with the periodic model retaining the dependence on I . Alongside patch Catalan numbers, counting appropriate maximal Deograms, I plan to study their q, t -refinements and characterize positive real loci by positivity of the variables in one seed. I also want a link attached to each $\mathring{R}_{f,\beta}$. In finite type, the graded Khovanov–Rozansky homology of positroid links recovers geometric point-count and q, t -data [6]. An affine analogue should recover affine Deodhar point counts and extend the connection in Theorem 3 to patches, while respecting the distinction between intrinsic torus structure and standard torus orbits.

My undergraduate research at Harvey Mudd College led to seven published papers. Since 2023, I have mentored eight students through the UCLA Directed Reading Program and now serve on its organizing committee. In Spring 2026, Olha Shevchenko and I co-mentored three students on a research project that led to a preprint [2]. I would like to continue mentoring undergraduates through directed reading and research projects.

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